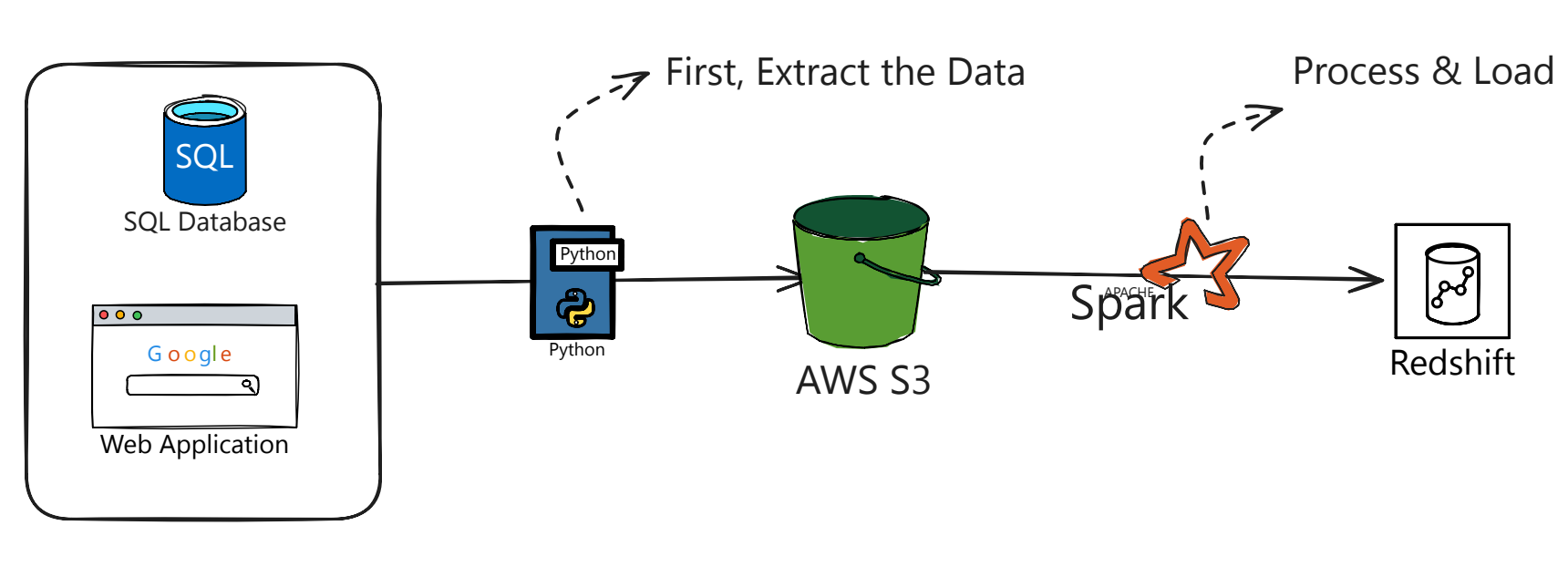
Airflow Masterclass

What is Airflow?

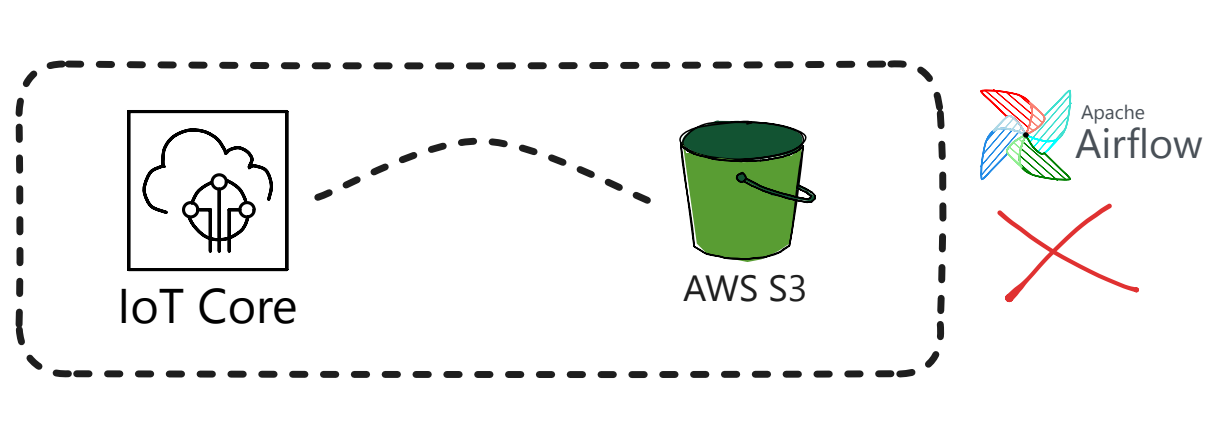
Airflow is an open source framework which can be used as an ORCHESTRATOR


SQL
SQL Database
Web Application
Python
First, Extract the Data
AWS S3
Spark
Process & Load Data
Redshift

What Airflow Is Not?

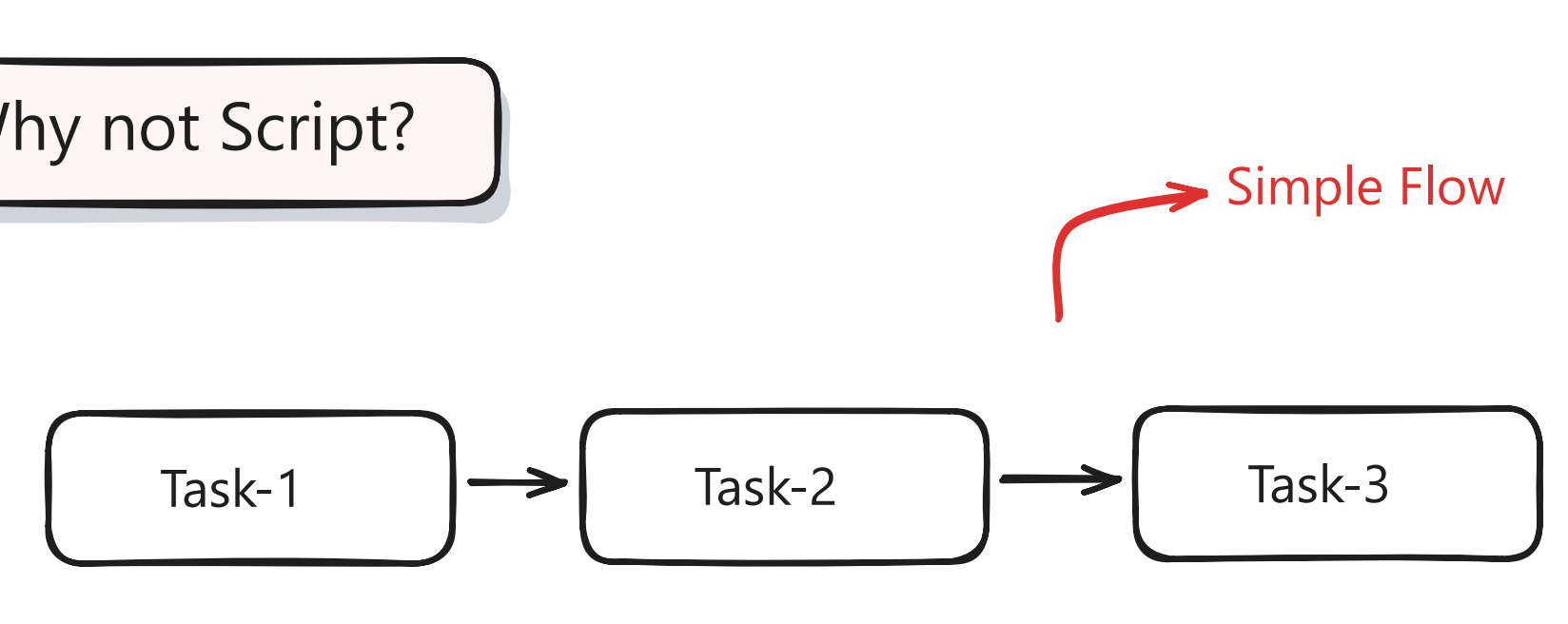
Airflow is not a DATA PROCESSING framework.

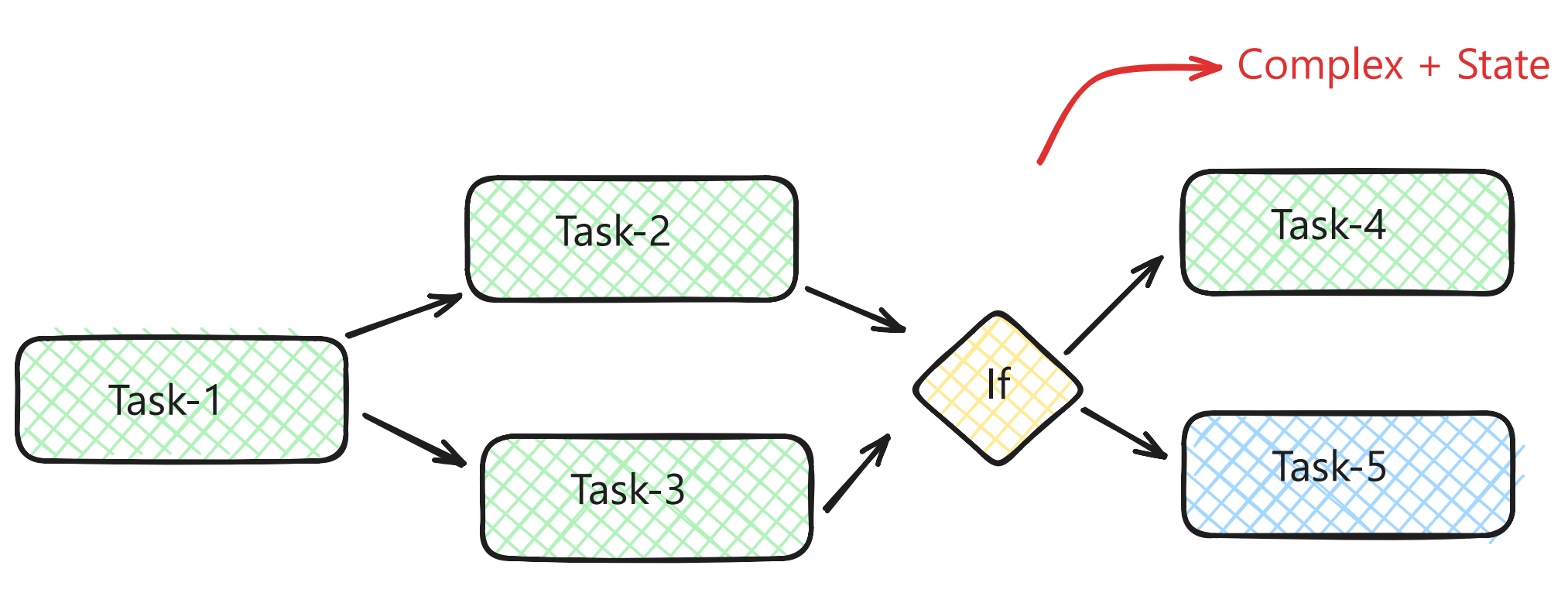
Airflow is not a REAL-TIME PROCESSING framework.


IoT Core
AWS S3
Apache Airflow

Airflow is not an ETL framework.

Why not Script?


Simple Flow
Task-1
Task-2
Task-3

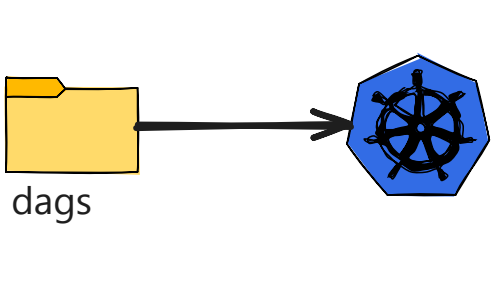

Complex + State
Task-1
Task-2
Task-3
If
Task-4
Task-5

Core Components of Airflow

Metadata DB

Airflow needs to store all the metadata about the DAGs.
For instance, DAG runs, schedule, status, task instance etc.

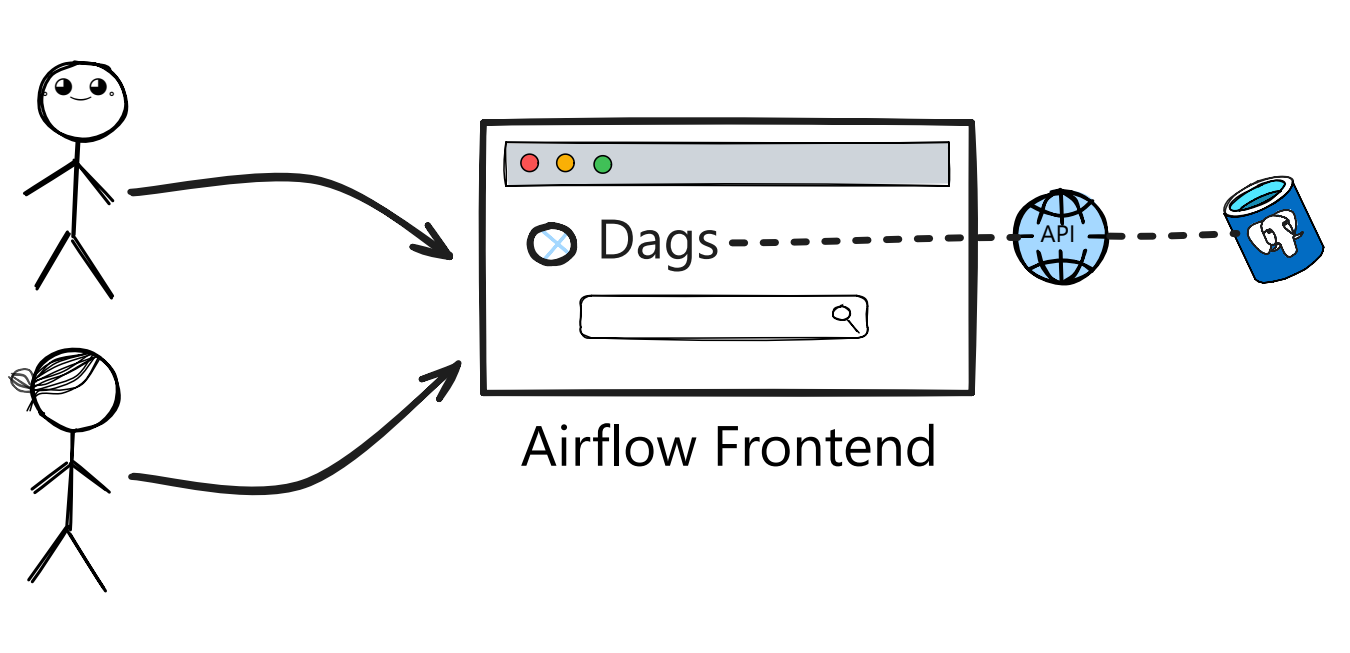
DAG File Processor



DAG File processor will go to your dags folder and will parse it and store the serialized DAG into the DB

{"xyz": "abc"}

API Server


Dags
Airflow Frontend

Scheduler

Scheduler is responsible for WHAT & WHEN tasks need to be executed.

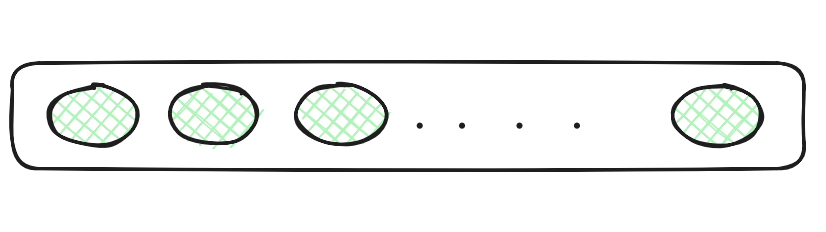
Executor

Executor will decide the HOW & WHERE tasks need to be run.

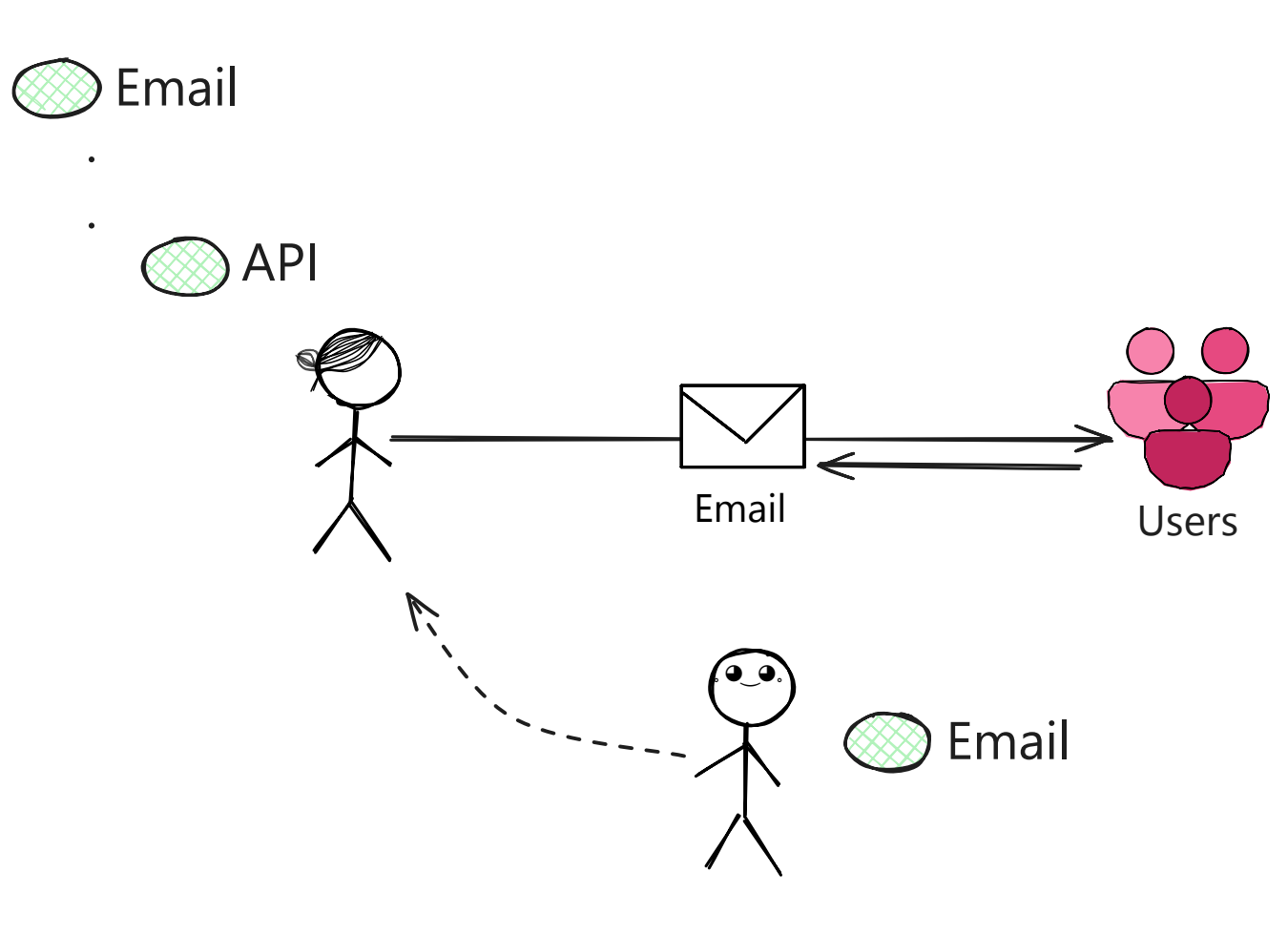
Workers

Workers are the real layer for execution.

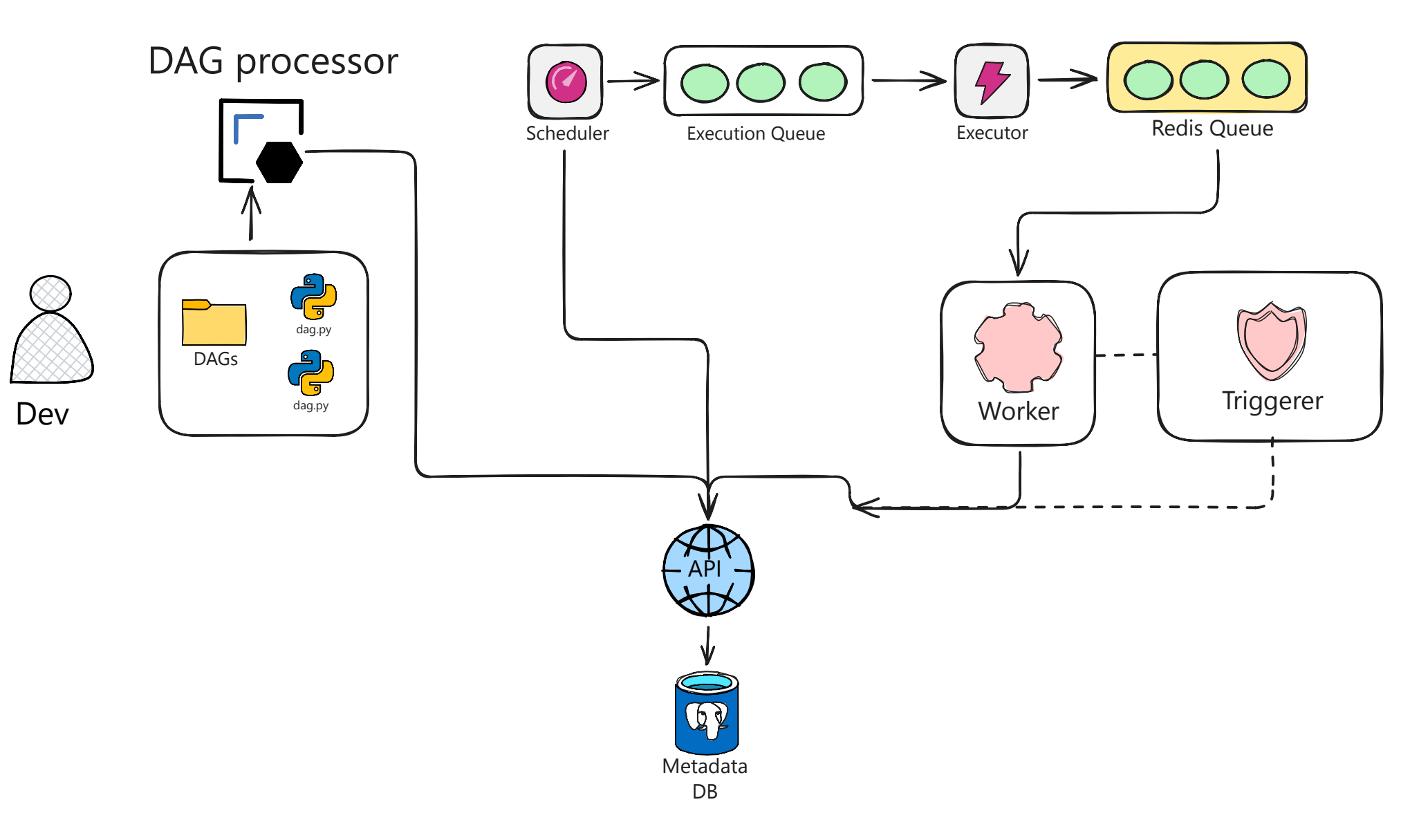
Queue



Triggerer


Email
API
Email
Users
Email

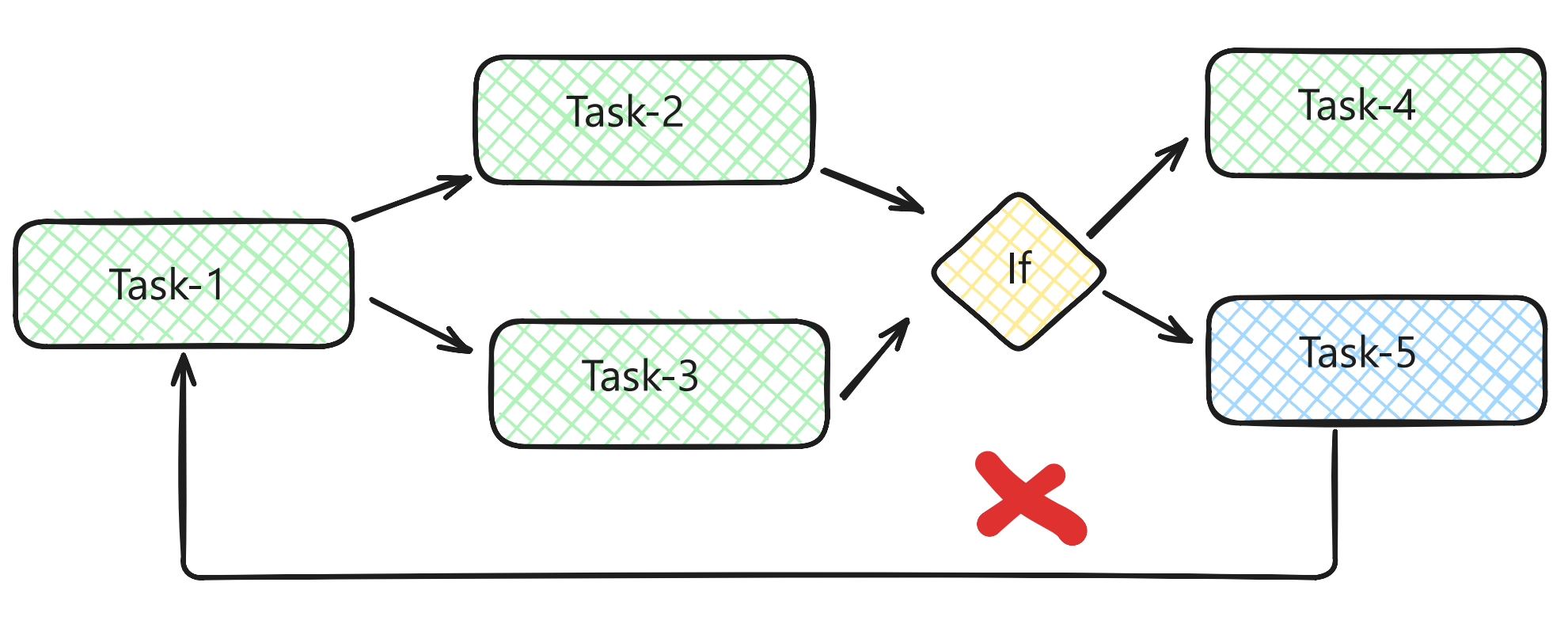
Airflow Architecture (Imp)

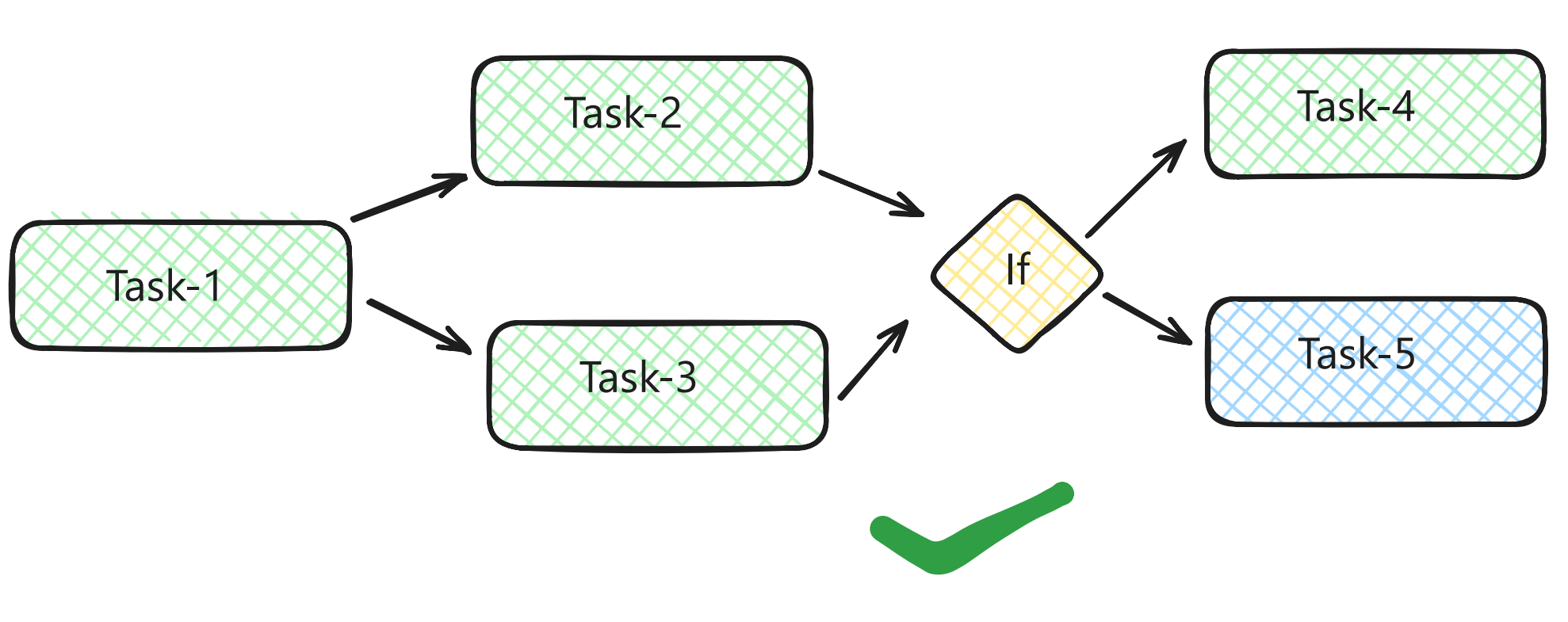

DAG processor
Dev
DAGs
dag.py
dag.py
Scheduler
Execution Queue
Executor
Redis Queue
Worker
Triggerer
API
Metadata DB

Core Concepts of Airflow

DAG

Directed Acyclic Graph


Task-1
Task-2
Task-3
If
Task-4
Task-5


Task-1
Task-2
Task-3
If
Task-4
Task-5

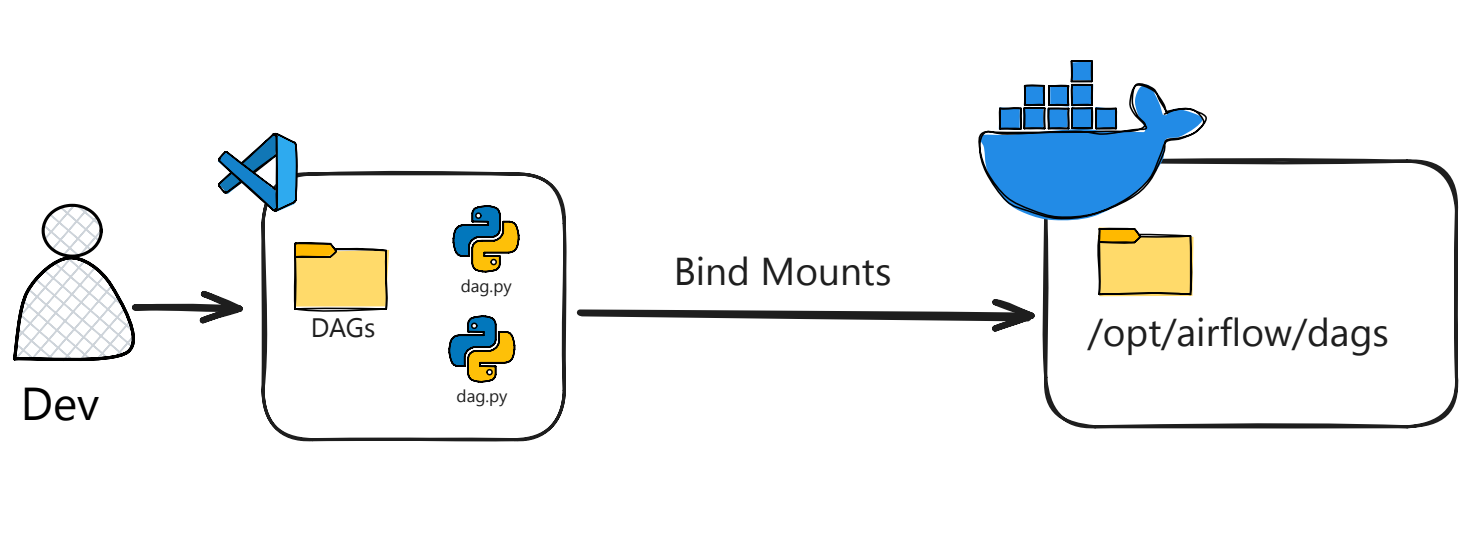
Task Instance

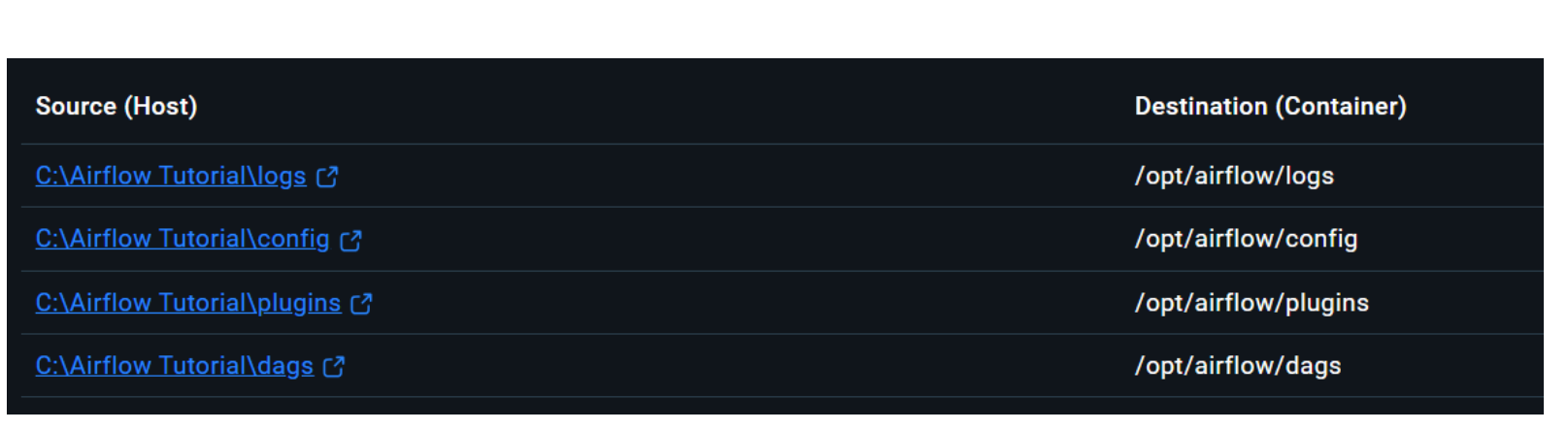
It's just a single unit of work.

Operator

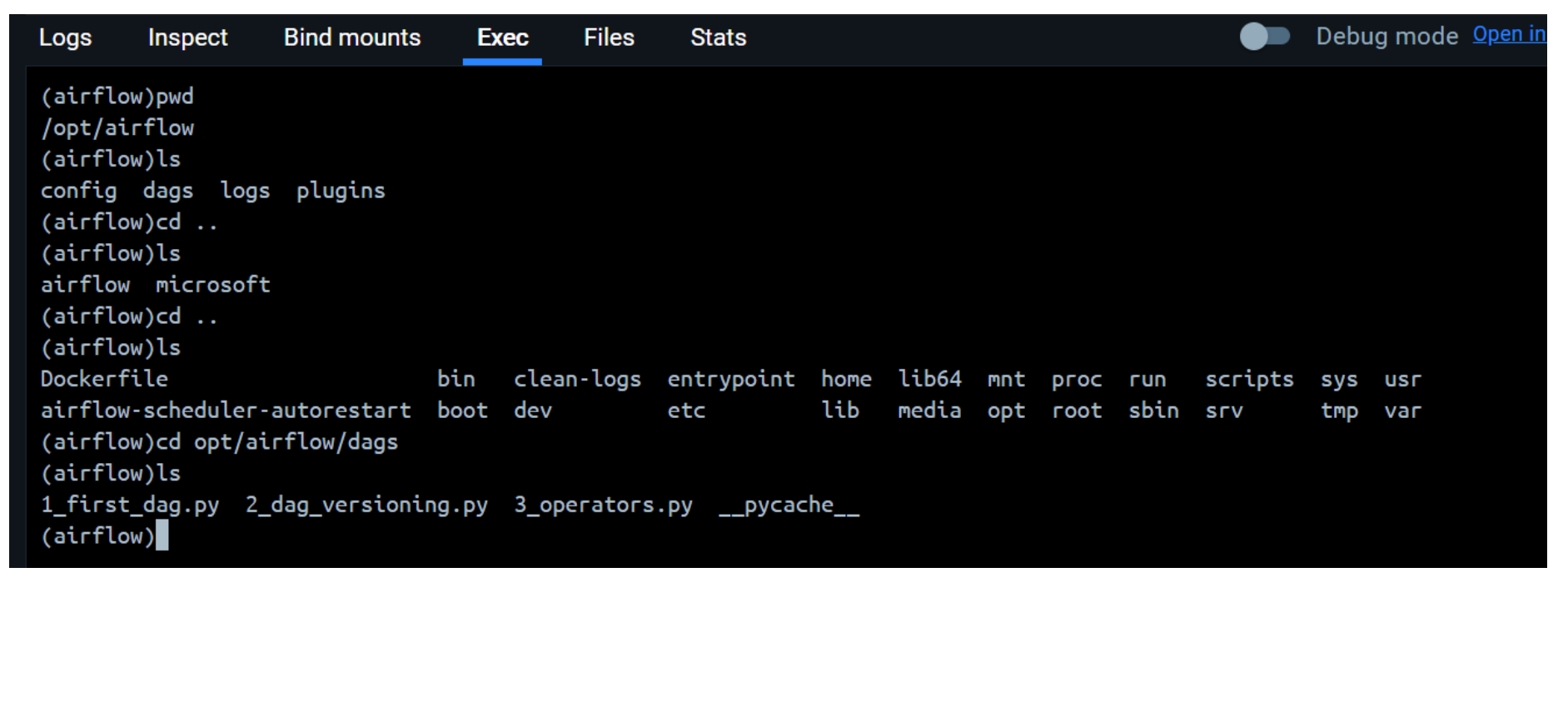
It's the pre-built template for our tasks.
It basically tells Airflow what type of task we are running.

How DAGs are synced?


Dev
DAGs
dag.py
dag.py
Bind Mounts
/opt/airflow/dags

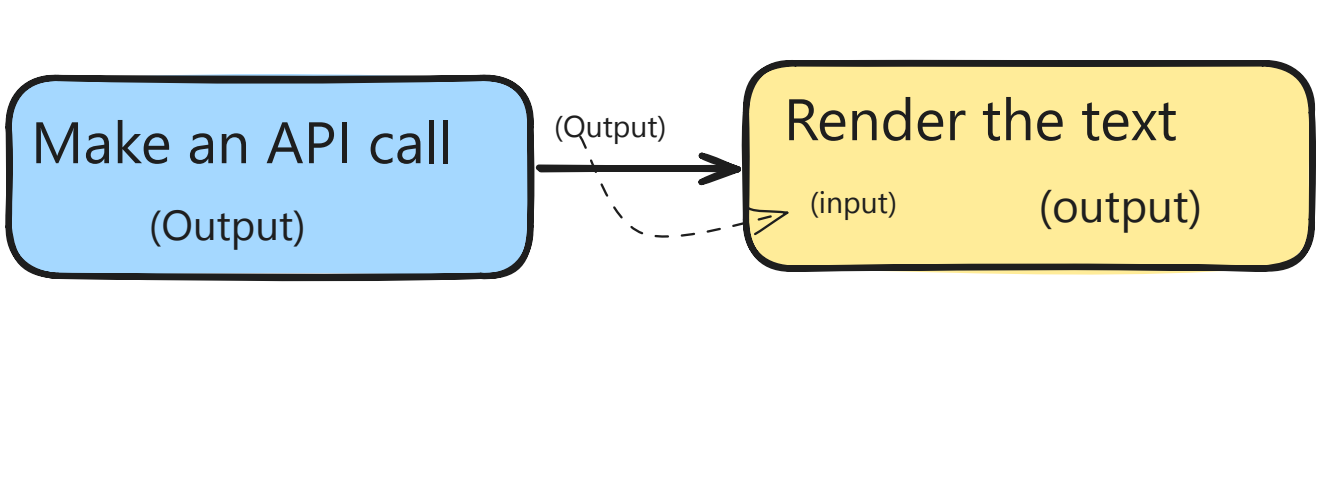


Source (Host)	Destination (Container)
C:\Airflow Tutorial\logs	/opt/airflow/logs
C:\Airflow Tutorial\config	/opt/airflow/config
C:\Airflow Tutorial\plugins	/opt/airflow/plugins
C:\Airflow Tutorial\dags	/opt/airflow/dags

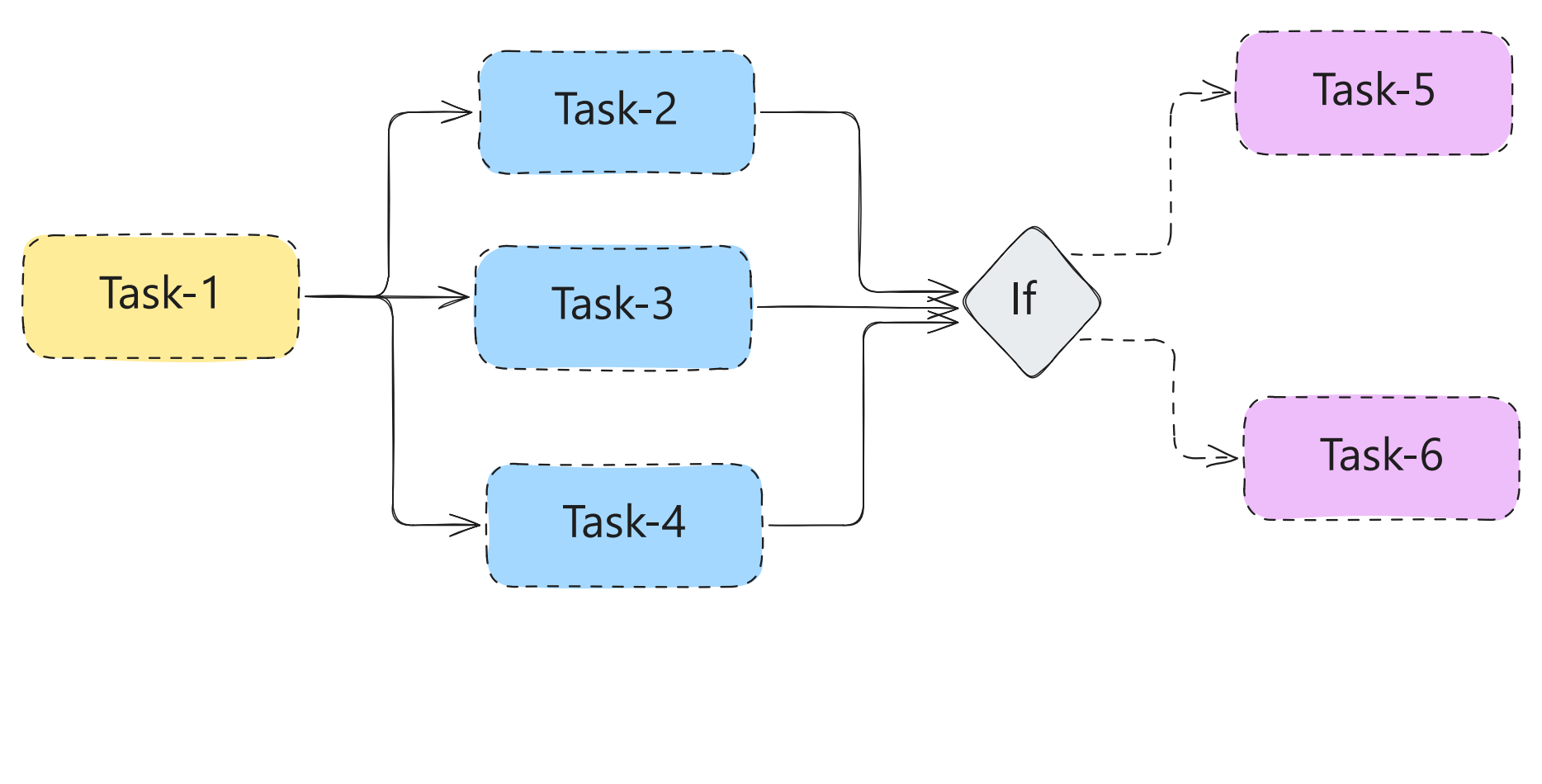

Logs
Inspect
Bind mounts
Exec
Files
Stats
Debug mode
Open in

```
(airflow)pwd
/opt/airflow
(airflow)ls
config  dags  logs  plugins
(airflow)cd ..
(airflow)ls
airflow  microsoft
(airflow)cd ..
(airflow)ls
Dockerfile                     bin   clean-logs  entrypoint  home  lib64  mnt  proc  run   scripts  sys  usr
airflow-scheduler-autorestart  boot  dev         etc         lib   media  opt  root  sbin  srv      tmp  var
(airflow)cd opt/airflow/dags
(airflow)ls
1_first_dag.py  2_dag_versioning.py  3_operators.py  __pycache__
(airflow)
```

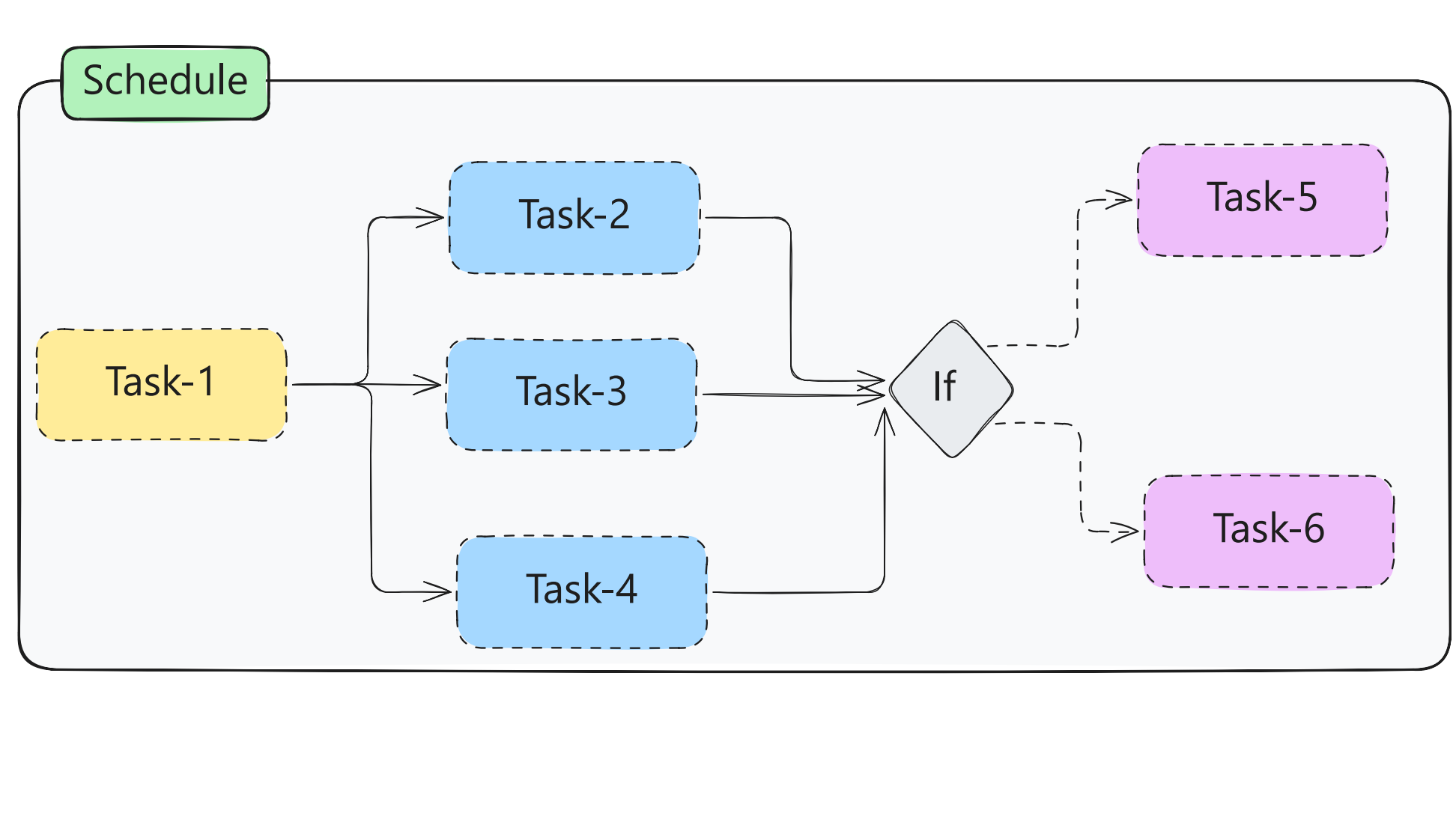
XCOMs In Airflow


Make an API call
(Output)
(Qutput)
Render the text
(input)
(output)

Parallel Tasks


Task-1
Task-2
Task-3
Task-4
If
Task-5
Task-6

Scheduling In Airflow


Schedule
Task-1
Task-2
Task-3
Task-4
If
Task-5
Task-6

start_date (Required)

end_date (optional)

catchup (optional)

It is used to tell Airflow to run the DAG for the missed intervals

schedule (Required)

Presets

@daily
@hourly
@weekly

Cron

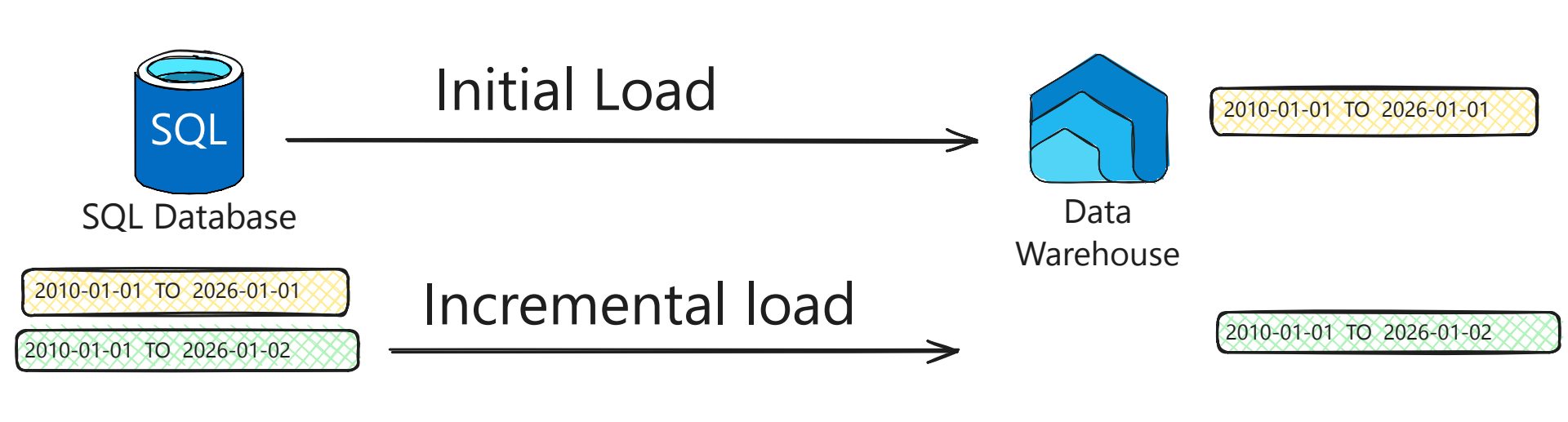
0 16 * * MON-FRI

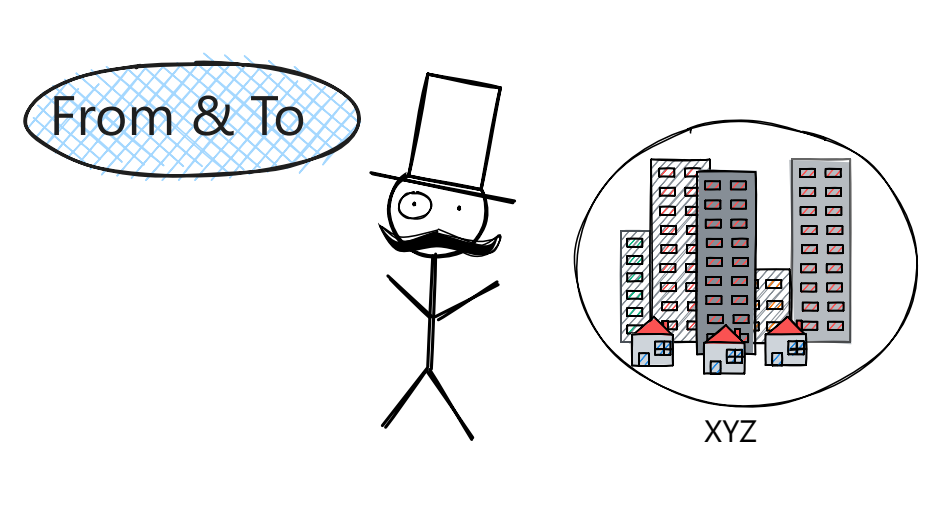
Delta

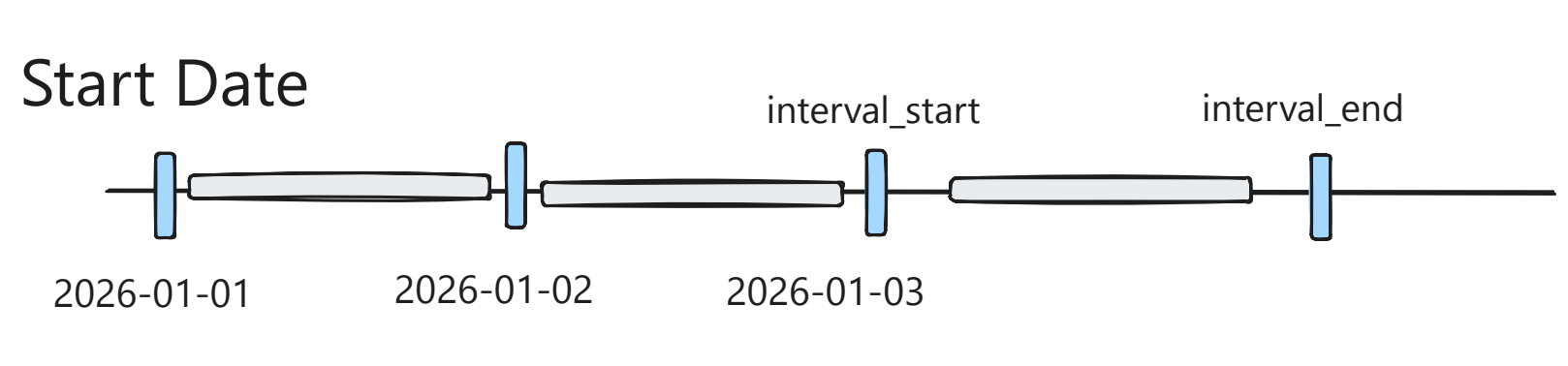
1,4,7,10,13,16,19,22,25,28,31

Feb
1,4,7,10,13,16,19,22,25,28,31
3,6,9,12,15,18,21,24,27

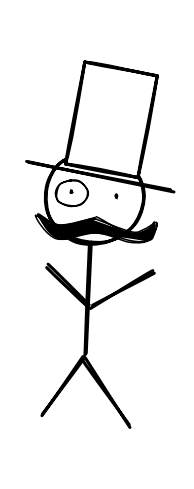
Incremental Load & Templating


SQL
SQL Database
2010-01-01 TO 2026-01-01
2010-01-01 TO 2026-01-02
Initial Load
Incremental load
Data Warehouse
2010-01-01 TO 2026-01-01
2010-01-01 TO 2026-01-02



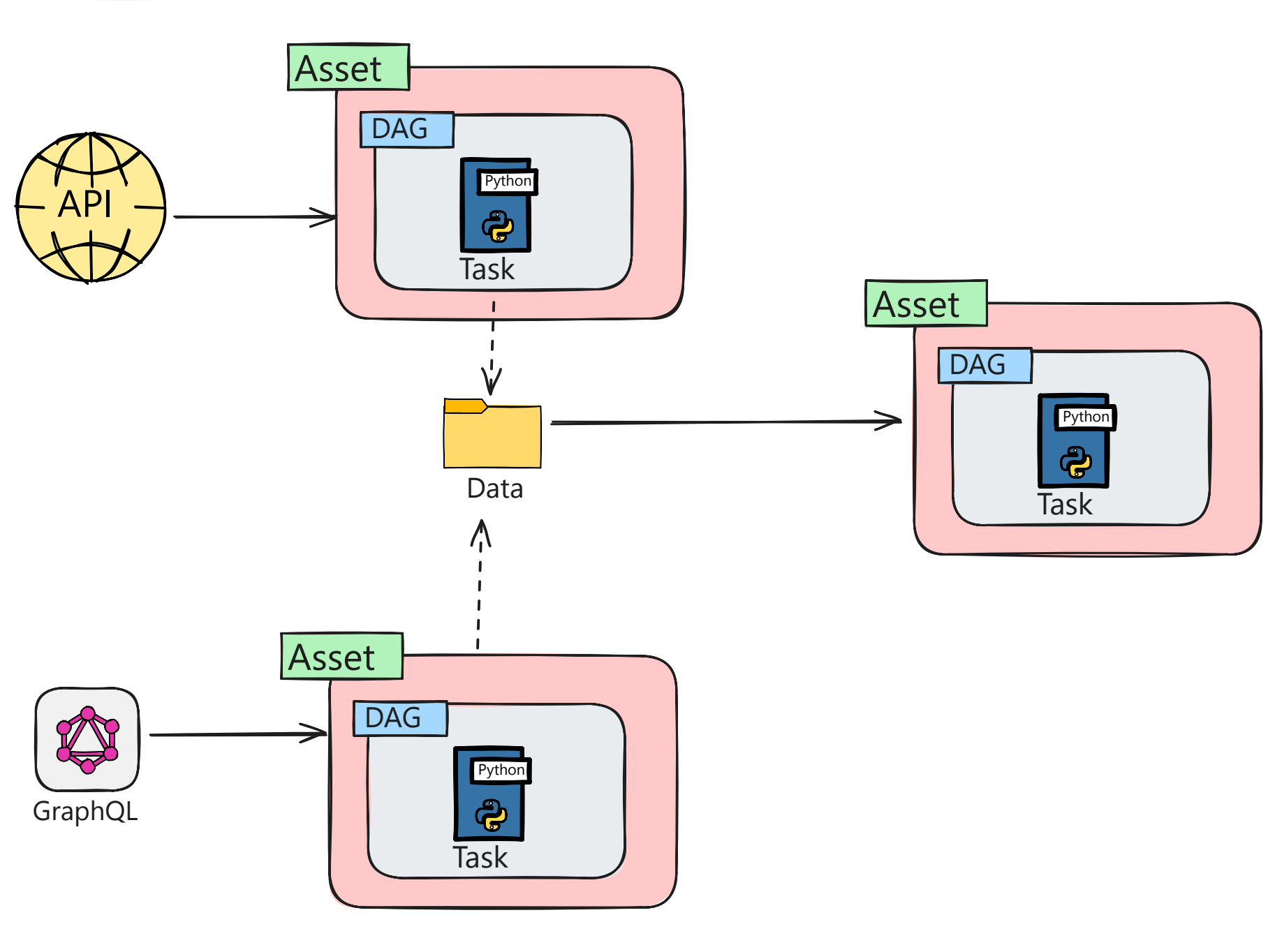

Start Date
interval_start
interval_end
2026-01-01
2026-01-02
2026-01-03

Special Schedules - Events

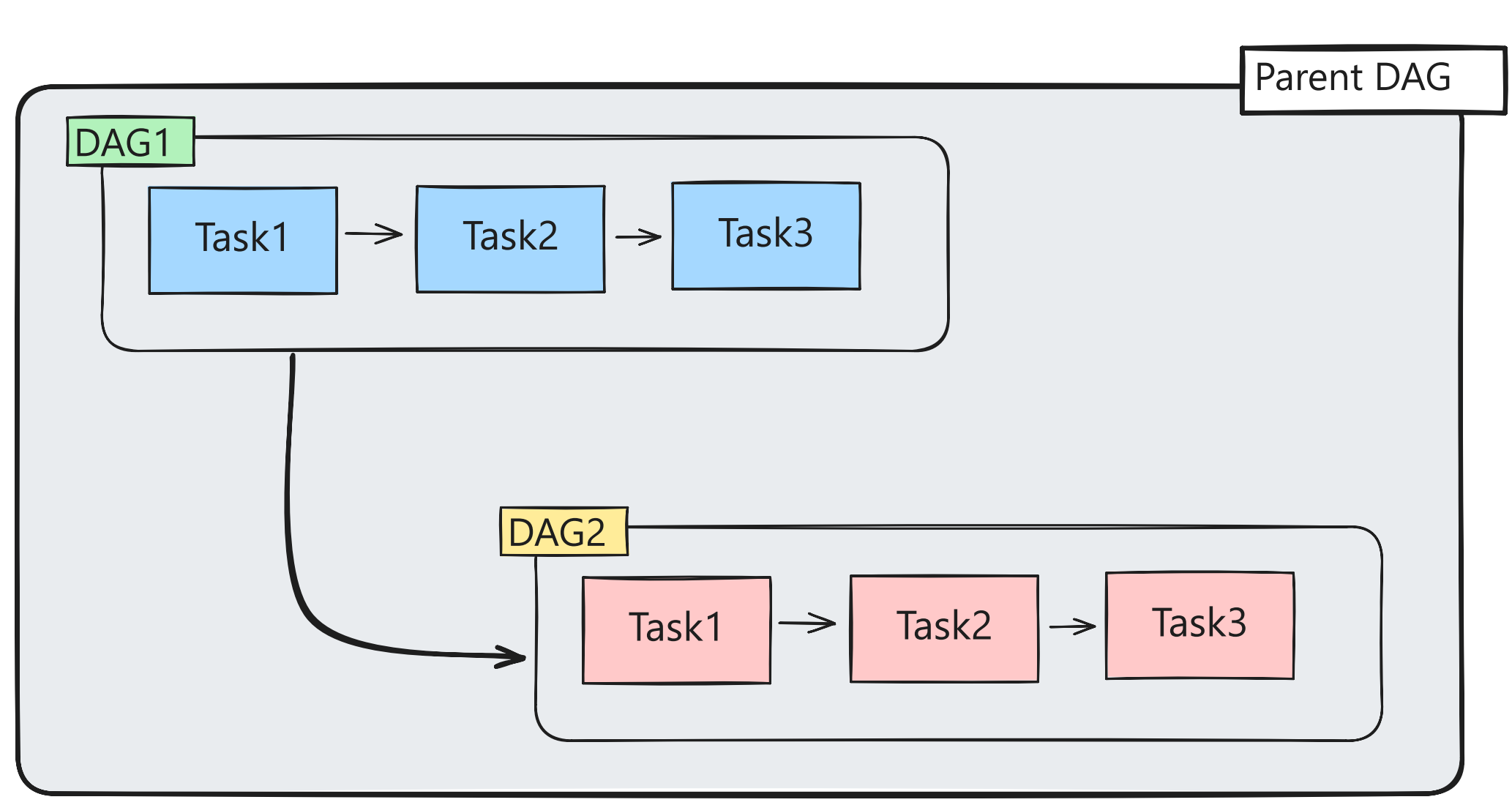


2026-01-01
2026-02-16
2026-06-01
2026-12-31

Assets


Asset
DAG
Python
Task
API
Data
Asset
DAG
Python
Task
Asset
DAG
Python
Task
GraphQL

DAGs Orchestration


Parent DAG
DAG1
Task1
Task2
Task3
DAG2
Task1
Task2
Task3